

Computational Astrophysics Exercises — Assignments Set 4

Integration

The error of an integration scheme is defined as

$$E[f] = I[f] - Q[f] \quad (1)$$

where $I[f]$ denotes the exact solution of the integral and $Q[f]$ the numerical solution for the integral of function $f(x)$ in the interval $[a, b]$.

The scheme for Simpson's integration reads

$$Q_2[f] = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]. \quad (2)$$

The error of an integration scheme with polynomials in the order of n is given for even n by

$$E_n[f] = \frac{K_n}{(n+2)!} f^{(n+2)}(\xi) \quad \text{with} \quad K_n = \int_a^b x W_n(x) dx, \quad (3)$$

where ξ is a point in $[a, b]$. The error polynomial $W_n(x)$ is given by

$$W_n(x) = \prod_{i=0}^n (x - x_i), \quad (4)$$

where x_i denote the nodes of the polynomial.

1. Simpson's rule — Error estimation

Show, that the integration error of Simpson's rule ($n=2$) is given by

$$E_2[f] = -\frac{f^{(4)}(\xi)}{90} \left(\frac{b-a}{2} \right)^5, \quad (5)$$

with the nodes $x_0 = a, x_1 = (b+a)/2, x_2 = b$.

Hint: Use the variable transformation

$$t = \frac{x-a}{b-a}, \quad (6)$$

that transforms the interval $[a, b]$ to $[0, 1]$.

(ob11 points)

2. Integration using Trapez, Simpson's and the adaptive Simpson's method

Develop a program to calculate the integral $I(f)$ of the function $f(x)$ over the interval $[a, b]$.

$$I(f) = \int_a^b f(x) dx. \quad (7)$$

Your program should implement the Trapez method, the Simpson's rule and the adaptive Simpson's method. The last method uses an estimate of the error using Simpson's rule. If the error exceeds a defined value, the algorithm subdivides the interval of integration in two and the adaptive Simpson's method is applied to each subinterval in a recursive manner, see, e.g., https://en.wikipedia.org/wiki/Adaptive_Simpson%27s_method.

Calculate the values for the Si and Fresnel integrals $Si(1)$ and $C(5)$

$$Si(x) = \int_0^x \frac{\sin(t)}{t} dt \quad \text{and} \quad C(x) = \int_0^x \cos\left(\frac{\pi}{2}t^2\right) dt, \quad (8)$$

with an accuracy of $\varepsilon = 10^{-8}$ using the adaptive Simpson's rule¹. Calculate the integral using the Trapez method and lower (slice the interval into 2) the step size until you have reached an accuracy of $\varepsilon = 10^{-8}$ using the following scheme:

1. Choose an initial number of steps $N_1 = 1$ and calculate the first approximation I_1 to the integral.
2. Double the number of steps and calculate an improved approximation I_2 .
3. Continue until the error between the two last steps is less than ε .

(Ob111 points)

in total

($\approx 10^4$ masses of Jupiter in units of $M_\odot$ points)

Deadline:

Fri, 29 May at 11:59am  via email (including code) to christoph.schaefer@uni-tuebingen.de.

¹You might want to compare your implementation's results to the scipy implementation of Simpson's rule <https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.simpson.html>